

St Catherine's School

Year: 12

Subject: Extension I Mathematics

Time allowed: 2 hours

(plus 5 mins reading time)

Date: August 2002

Exam number: _____

Directions to candidates:

- All questions are to be attempted.
- All questions are of equal value.
- All necessary **working** must be shown in every question.
- Full marks may not be awarded for careless or badly arranged work.
- Approved calculators and geometrical instruments are required.

- Each **section** should be started on a **new booklet**.
- Hand in your work in **3 bundles**:

Section A Questions 1 and 2.

Section B Questions. 3 and 4

Section C Questions. 5, 6 and 7.

TEACHER'S USE ONLY Total Marks

A

B

C

TOTAL

Section A

Question 1

- a) Use the table of standard integrals to find the exact value of $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$ 2
- b) Find $\frac{d}{dx} \sin^{-1} \sqrt{1-x}$ 2
- c) Evaluate $\sum_{n=3}^7 (3n-1)$ 1
- d) Let A be the point (x_1, y_1) and let B be the point (x_2, y_2) . Find the coordinates of the point P, which divides the interval AB externally in the ratio 3:2. 2
- e) Is $x-2$ a factor of $x^3 + 3x - 14$? Give reason for your answer. 2
- f) Use the substitution $u = x^2 - 1$ to evaluate $3 \int_1^2 x \sqrt{x^2 - 1} dx$ 3

Question 2

- a) Let $f(x) = 2x^2 + x$. Use the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find the derivative of $y = f(x)$. 2
- b) i) Find $\int \frac{e^{2x}}{3 + e^{2x}} dx$ 1
- ii) Evaluate $\int_0^{\frac{\pi}{2}} \sin^2(\frac{1}{2}x) dx$ 3
- c) i) Write down the expansion of $\tan(A+B)$ 1
- ii) Hence find the value of $\tan 105^\circ$ in simplest surd form. 2
- d) Solve for x; $\frac{4}{5-x} \geq 1$ 3

Section B (Start a new booklet)

Question 3

- a) Write the expansion of $(2x - y)^5$ 2
- b) Find the term independent of x in the binomial expansion $(x^2 + \frac{2}{x})^6$ 3
- c) The function $f(x) = x - 2\sin x$ has a zero near $x = 1.7$. Use one application of Newton's method to find a second approximation to the zero. Write your answer correct to three significant figures. 3
- d) A smooth piece of ice is projected up a smooth inclined (sloping) surface. Its distance x in metres up the surface is at time t seconds is $x = 6t - t^2$. 4
- i) Find velocity, v , and acceleration, $\ddot{x}$.
 - ii) In which direction is the ice moving and in which direction is acceleration when $t = 2$
 - iii) Use your answer from (ii) to explain whether the piece of ice is increasing in speed or decreasing in speed when $t = 2$.
 - iv) Find when and where the piece of ice is stationary.

Question 4

- a) i) Given that $x^2 + 4x + 13 \equiv (x + a)^2 + b^2$, find the values of a and b . 1
- ii) Use the result of (i) to find $\int_{-2}^1 \frac{1}{x^2 + 4x + 13} dx$ 2
- b) The volume, V , of a sphere of radius r mm is increasing at a constant rate of 200mm^3 per second. 4
- i) Find $\frac{dr}{dt}$ when $r = 50$.
 - ii) Determine the rate of increase of the surface area, S of the sphere when the radius is 50 mm.
- $$\left(V = \frac{4}{3}\pi r^3 \quad S = 4\pi r^2 \right)$$
- c) A particle, whose displacement is x , moves in simple harmonic motion. 5
- Find x as a function t if:
- $$\ddot{x} = -16x \text{ and } x = \sqrt{3} \text{ and } \dot{x} = 12 \text{ when } t = 0.$$

Section C (Start a new booklet)

Question 5

- a) $P(2p, p^2)$ and $Q(2q, q^2)$ are two points on the parabola $x^2 = 4y$. The variable chord PQ is such that it is always parallel to the line $y = x$.
- (i) Find the gradient of PQ and hence show that $p + q = 2$ 1
- ii) Given that the equation of the *normal* at P is $x + py = 2p + p^3$. Write down the equation of the *normal* at Q, and hence find the coordinates of the point of intersection R, of these normals. 3
- iii) Prove that the locus of R is the straight line $x - 2y + 12 = 0$. 2
- b) A garden sprinkler is positioned at the centre of a large, flat lawn. Water droplets are projected from the sprinkler at a fixed speed of 20ms and at an angle, ϑ , above the horizontal. The acceleration due to gravity is 10 ms^{-2} .
- i) Use integration to show that the horizontal displacement x metres and the vertical displacement y metres of the water droplets is after time t seconds are given by 2
- $$x = 20t \cos \vartheta \quad \text{and} \quad y = 20t \sin \vartheta - 5t^2$$
- ii) Show that the horizontal range, R , of the water droplets is given by $R = 40 \sin 2\vartheta$. 2
- (iii) The garden sprinkler rotates in a circle to water the lawn. If the angle of projection varies between 15° and 45° above the horizontal, find the exact area of that part of the lawn that can be watered in this way. 2

Question 6

- a) The acceleration of an object is given by $a = 12e^{2x} \text{ m/s}^2$. If the object leaves the origin with a velocity of $2\sqrt{3} \text{ m/s}$ and its velocity is always positive, find the displacement when the velocity is 5 m/s , correct to two decimal places. 4
- b) i) Find the general solution to the equation $\sin 2x = 2 \sin^2 x$. 3
- ii) Show that if $0 < x < \frac{\pi}{4}$, then $\sin 2x > 2 \sin^2 x$ 2
- iii) Find the area enclosed between the curves $y = \sin 2x$ and $y = 2 \sin^2 x$ for $0 \leq x \leq \frac{\pi}{4}$ 3

Please turn over for question 7

Question 7

- a) In a flock of 1000 chickens, the number P , infected with a disease at time t years is given by

$$P = \frac{1000}{1 + ke^{-1000t}}$$
 where k is a constant. 1
- i) Show that, eventually, all the chickens will be infected. 1
- ii) Suppose that when time $t = 0$, exactly one chicken was infected. After how many days will 500 chickens be infected? 3
- b) A particle is moving in a straight line. After time, t seconds it has displacement x metre from a fixed point O on the line, velocity, $v \text{ ms}^{-1}$ given by $v = \frac{1-x^2}{2}$ and acceleration, $a \text{ ms}^{-2}$. Initially the particle is at O . 1
- i) Find an expression for acceleration in terms of x . 1
- ii) Show that $\frac{2}{1-x^2} = \frac{1}{1+x} + \frac{1}{1-x}$ and hence find an expression for x in terms of t . 3
- iii) Describe the initial conditions for displacement, velocity and acceleration and hence describe the motion of the particle, explaining whether it moves to the left or the right of O and whether it slows down or speeds up. 3
- iv) What is the limiting position of the particle? 1

End of examination

Question 1

a)

$$\int_0^1 \frac{dx}{\sqrt{4-x^2}} = \left[\sin^{-1} \left(\frac{x}{2} \right) \right]_0^1$$

$$= \sin^{-1} \frac{1}{2} - \sin^{-1} 0$$

$$= \frac{\pi}{6} - 0$$

$$= \frac{\pi}{6}$$

(2)

b)

$$\frac{d}{dx} \sin^{-1} \sqrt{1-x} =$$

$$\text{let } u = \sqrt{1-x} = (1-x)^{1/2}$$

$$\frac{du}{dx} = -\frac{1}{2}(1-x)^{-1/2}$$

$$\text{let } y = \sin^{-1} u$$

$$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$$

$$\frac{dy}{dx} = \frac{du}{dx} \times \frac{dy}{du}$$

$$= \frac{-1}{2\sqrt{1-x}} \times \frac{1}{\sqrt{1-(1-x)}}$$

$$= \frac{-1}{2\sqrt{1-x}} \times \frac{1}{\sqrt{x}}$$

(2)

$$c) \sum_{n=3}^7 (3n-1) = 8+11+14+17+20$$

$$= 70$$

(1)

$$d) A(6,2) B(4,7) \quad m: n = -3:2$$

$$x = \frac{mx_2 + nx_1}{m+n} \quad y = \frac{my_2 + ny_1}{m+n}$$

$$= \frac{3(4) - 2(-6)}{3-2} = \frac{3(7) - 2(2)}{3-1}$$

(2)

$$= 24 \quad = 17$$

$$\text{ie. } (24, 17)$$

e) If $(x-2)$ is a factor of $x^3+3x-14$
then when $x=2$ $x^3+3x-14=0$

$$P(2) = (2)^3 + 3(2) - 14$$

$$= 8 + 6 - 14$$

$$= 0$$

(2)

$\therefore (x-2)$ is a factor by the factor theorem

$$f) \int_0^2 x \sqrt{x^2-1} \, dx \quad u = x^2-1$$

$$\frac{du}{dx} = 2x$$

$$du = 2x \, dx$$

$$\frac{1}{2} du = x \, dx$$

$$\text{at } x=2 \quad u=3$$

$$x=1 \quad u=0$$

$$= \frac{3}{2} \int_0^3 u^{1/2} \, du$$

$$= \frac{3}{2} \left[\frac{2}{3} u^{3/2} \right]_0^3$$

$$= \left[u^{3/2} \right]_0^3$$

$$= (3)^{3/2}$$

$$= \sqrt{27}$$

$$= 3\sqrt{3}$$

(3)

Stephanie Zimm.

Question Two

$$(a) f(x) = 2x^2 + x$$

$$f(x+h) = 2(x+h)^2 + (x+h)$$

$$= 2(x^2 + 2xh + h^2) + (x+h)$$

$$= 2x^2 + 4xh + 2h^2 + x + h$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 + x + h - (2x^2 + x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 + h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(4x + 2h + 1)}{h}$$

$$= \lim_{h \rightarrow 0} 4x + 2h + 1$$

$$= 4x + 1$$

(2)

$$(b) i) \int \frac{e^{2x}}{3+e^{2x}} \, dx$$

$$= \frac{1}{2} \ln(3+e^{2x}) + C$$

(1)

$$ii) \int_0^{\pi/2} \sin^2 \left(\frac{1}{2}x \right) \, dx$$

$$\frac{1}{2} \int 1 - \cos x \, dx$$

$$= \frac{1}{2} \left[x - \sin x \right]_0^{\pi/2}$$

$$= \frac{1}{2} \left[\frac{\pi}{2} - 1 \right]$$

$$ci) \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$ii) \tan(105) = \tan(60+45)$$

$$= \frac{\tan 60 + \tan 45}{1 - \tan 60 \cdot \tan 45}$$

$$= \frac{\sqrt{3} + 1}{1 - \sqrt{3} \cdot 1} \times \frac{1+\sqrt{3}}{1+\sqrt{3}} = \frac{4+2\sqrt{3}}{-2}$$

$$d) \frac{4}{5-x} > 1$$

METHOD ONE: squaring Both sides

$$\frac{4}{5-x} (5-x)^2 > (5-x)^2, x \neq 5$$

$$4(5-x) > 25 - 10x + x^2$$

$$20 - 4x > 25 - 10x + x^2$$

$$0 > x^2 - 6x + 5$$

$$0 > (x-5)(x-1)$$



$$1 < x < 5$$

(3)

METHOD TWO Critical Points Method.

$$\frac{4}{5-x} \geq 1$$

$$x \neq 5$$

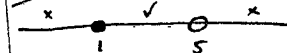
Solve equality

$$\frac{4}{5-x} = 1$$

$$4 = 5-x$$

$$x = 1$$

Test



$$5 \leq x < 5$$

Question 3

$$a) (2x-y)^5 = {}^5C_0 (-y)^0 (2x)^5 + {}^5C_1 (-y)^1 (2x)^4 + {}^5C_2 (-y)^2 (2x)^3 + {}^5C_3 (-y)^3 (2x)^2 + {}^5C_4 (-y)^4 (2x)^1 + {}^5C_5 (-y)^5 (2x)^0$$

$$= -y^5 + 5y^4 \times 2x + 10(-y)^2 (4x^3) + 10y^3 \cdot 8x^2 + 5(-y)(16x^4) + 1(32x^5)$$

$$\textcircled{2} = -y^5 + 10xy^4 - 40x^2y^3 + 80x^3y^2 - 80x^4y + 32x^5$$

$$b) T_{k+1} = {}^nC_k \left(\frac{2}{x}\right)^k (x^2)^{n-k} \quad n=6$$

$$= {}^6C_k \left(\frac{2}{x}\right)^k (x^2)^{6-k}$$

$$= {}^6C_k (2^k) (x^{-k}) (x^2)^{6-k}$$

$$= {}^6C_k 2^k x^{12-3k}$$

now $x^{12-3k} = x^0$

$\therefore k=4$

$\therefore T_5$ is the independent term $\frac{1}{x^2}$

$$T_5 = {}^6C_4 \left(\frac{2}{x}\right)^4 (x^2)^2 \quad \textcircled{1}$$

$$= 15 \cdot \frac{16}{x^4} \cdot x^4$$

$$= 15 \times 16$$

$$= \underline{240}$$

$\textcircled{3}$

$\frac{1}{2}$ marks

c) $f(x) = x - 2\sin x$ has a root near 1.7

$f'(x) = 1 - 2\cos x \quad \left(\frac{1}{2}\right)$

$$\textcircled{3} \quad x = 1.7 - \frac{f(1.7)}{f'(1.7)} \quad \left(\frac{1}{2}\right)$$

$$= 1.7 + \frac{1.64...}{0.99...} \quad \left\{ \text{if in degrees!} \right\}$$

$$= 3.34 \quad \left(\frac{1}{2}\right)$$

$$= \sqrt{1.92527...} = 1.93 \quad \left(\frac{1}{2}\right)$$

1.7 is in radians

$$\textcircled{1} \quad f(1.7) = -0.28332962$$

$$\textcircled{1} \quad f'(1.7) = 1.257688989$$

$\left(\frac{1}{2}\right)$ for rads

page 3

$$3d) \quad x = 6t - t^2$$

$$\textcircled{1} \quad i) \quad \dot{x} = 6 - 2t$$

$$\ddot{x} = -2$$

$$ii) \quad \text{when } t=2, \quad \dot{x} = 6 - 2(2)$$

$$= 2$$

$\therefore$ it is moving to the right at 2 m/s

$\textcircled{1}$

$$\text{when } t=2, \quad \ddot{x} = -2$$

acceleration is in the neg direction (constantly)

iii) $\therefore$ the $\dot{x}$ is decreasing in speed since $\dot{x}$ and $\ddot{x}$ are in opposition

$\textcircled{1}$

$$iv) \quad \text{stat when } \dot{x} = 0$$

$$0 = 6 - 2t$$

$$6 = 2t$$

$$t = 3$$

$$x = 6(3) - (3)^2$$

$$= 18 - 9$$

$\therefore$ the ice is stationary

when $t=3$ at 9m up the slope.

$\textcircled{1}$

page 4

Question 4

a) $x^2 + 4x + 13 = (x+2)^2 + 9$

$(x^2 + 4x + 4) + 9 = (x+2)^2 + 9$

① $(x+2)^2 + 9 = (x+2)^2 + 9$

ii) $\int \frac{dx}{(x+2)^2 + 9}$

$= \int \frac{1}{3} \tan^{-1} \frac{x+2}{3} \Big|_{-2}^1$

$= \frac{1}{3} [\tan^{-1}(1) - \tan^{-1}(0)]$

$= \frac{1}{3} (\frac{\pi}{4})$

$= \frac{\pi}{12}$

②

b) i) $V = \frac{4}{3} \pi r^3$

$\frac{dV}{dr} = 4\pi r^2$

$\frac{dV}{dt} = 200 \text{ mm}^3/\text{s}$

$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$

② $200 = 4\pi r^2 \times \frac{dr}{dt}$

when $r = 50$

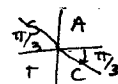
$200 = 4\pi (50)^2 \times \frac{dr}{dt}$

$\frac{200}{4\pi (2500)} = \frac{dr}{dt}$

$\frac{200}{10000\pi} = \frac{dr}{dt}$

$\frac{1}{50\pi} = \frac{dr}{dt}$

②



b) ii) $S = 4\pi r^2$

$\frac{dS}{dr} = 8\pi r$

$\frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt}$

$= 8\pi r \times \frac{1}{50\pi}$

$= \frac{8 \times 50}{50}$

$= 8 \text{ mm}^2/\text{sec}$

②

c) $\ddot{x} = -16x$

at $t = 0$
 $x = \sqrt{3}$ $\dot{x} = 12$

$x = a \cos(nt + \alpha)$

$\dot{x} = -na \sin(nt + \alpha)$

$\ddot{x} = -n^2 a \cos(nt + \alpha)$

at $t = 0$ $x = \sqrt{3}$ $\dot{x} = 12$

① $\sqrt{3} = a \cos \alpha$

② $12 = -na \sin \alpha$

but $n = 4\sqrt{2}$ since $\ddot{x} = -n^2 x$

③ $12 = -4a \sin \alpha$

solving ① & ③ simult

$\frac{\sqrt{3}}{3} = \frac{a \cos \alpha}{-a \sin \alpha}$

$\frac{1}{\sqrt{3}} = -\frac{a \sin \alpha}{a \cos \alpha}$

$-\frac{1}{\sqrt{3}} = \tan \alpha$

$\alpha = \frac{\pi}{3}$

$\alpha = \frac{\pi}{3}$

$a = 2\sqrt{3}$

$x = 2\sqrt{3} \cos(4t - \frac{\pi}{3})$

also $x = 2\sqrt{3} \sin(4t + \frac{\pi}{6})$

Question 5

4r 12, Ext 1, Trial 2002

a) i) $M_{PA} = +1$ since $PQ \parallel y = x$

$M_{PA} = \frac{p^2 - q^2}{2p - 2q}$

$= \frac{(p-q)(p+q)}{2(p-q)}$

$= \frac{p+q}{2}$

$1 = \frac{p+q}{2}$

① $\therefore p+q = 2$

b) ii) Normal at P

$x + py = 2p + p^3$

Normal at Q

$x + qy = 2q + q^3$

① - ②

$py - qy = 2p - 2q + p^3 - q^3$

$y(p-q) = 2(p-q) + (p-q)(p^2 + pq + q^2)$

$y = p^2 + pq + q^2 + 2$

$x = (p+q)^2 - pq + 2 = 6 - pq$

subst y coord into ①

$x + p(p^2 + pq + q^2 + 2) = 2p + p^3$

$x + p^3 + p^2q + pq^2 + 2p = 2p + p^3$

$x + pq^2 + pq^2 = 0$

$x = -pq(p+q)$ but $p+q = 2$

$x = -2pq$

$\therefore R(-2pq, 6-pq)$

iii) $x = -2pq$

$y = 6-pq$

$p = \frac{x}{-2q}$

$y = 6 - q(\frac{x}{-2q})$

$y = 6 + \frac{x}{2} \rightarrow x + 2y + 12 = 0$

Finding gradient $\frac{1}{2}$
equating to 1 $\frac{1}{2}$

$x = -pq(p+q)$
 $y = p^2 + pq + q^2 + 2$

$y = -2pq$

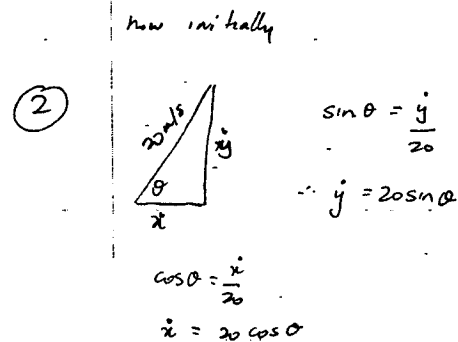
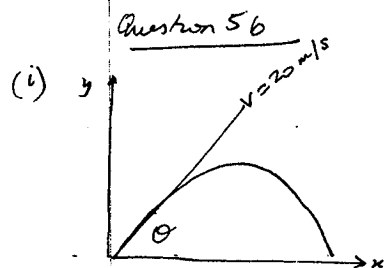
$x =$

manipulates to
for y is 1 mark.

writing $x =$

$y =$

using $p+q = 2$



horizontal motion	vertical motion
$\ddot{x} = 0$	$\ddot{y} = -10$
$\dot{x} = c_1$	$\dot{y} = -10t + c_2$
at $t=0$ $\dot{x} = 20 \cos \theta$	now at $t=0$ $\dot{y} = 20 \sin \theta$
$\therefore \dot{x} = 20 \cos \theta$	
$x = 20 \cos \theta t + c_2$	$\dot{y} = -10t + 20 \sin \theta$
at $t=0$ $x=0$	$y = -5t^2 + 20 \sin \theta t + c_3$
$0 = [20 \cos \theta](0) + c_2$	now at $t=0$ $y=0$
$c_2 = 0$	$0 = c_3$
$\therefore x = 20 \cos \theta t$	$\therefore y = -5t^2 + 20 \sin \theta t$

(ii) Range of x
 i.e. find x when $y=0$
 or find t when $y=0$
 then sub into x

at $y=0$
 $0 = -5t^2 + 20t \sin \theta$
 $50 = t(20 \sin \theta - 5t)$
 $0 = 5t(4 \sin \theta - t)$
 $\therefore t=0$ or $t = 4 \sin \theta$

at $t = 4 \sin \theta$
 $x = 20 \cos \theta \times 4 \sin \theta$
 $x = 80 \sin \theta \cos \theta$
 $= 40(2 \sin \theta \cos \theta)$
 $= 40 \sin 2\theta$

iii) $A = \pi(R^2 - r^2)$
 $= \pi(40^2 - 20^2)$
 $= \pi(1600 - 400)$
 $= 1200\pi \text{ sq m}$

at $\theta = 45^\circ$ $x = 40 \sin 90^\circ = 40$ i.e. $R = 40$
 at $\theta = 15^\circ$ $x = 40 \sin 30^\circ = 20$ i.e. $r = 20$

6a) $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 12e^{2x}$

$\frac{1}{2} v^2 = 6e^{2x} + C$
 $v^2 = 12e^{2x} + C$

when $x=0$ $v = 2\sqrt{3}$

$(2\sqrt{3})^2 = 12e^{2(0)} + C$
 $12 = 12e^{2(0)} + C$
 $12 = 12 + C$
 $C = 0$

$v^2 = 12e^{2x}$ ①

$v = \pm \sqrt{12e^{2x}}$

but $v > 0$ (since $x \geq 0$, $v > 0$)

$\therefore v = 2\sqrt{3}e^x$
 $= 2\sqrt{3}e^x$ (2dp) ①

$\frac{25}{12} = e^{2x}$

$2x = \ln \frac{25}{12}$

$x = \frac{1}{2} \ln \frac{25}{12}$

when $v=5$ find x

$5 = 2\sqrt{3}e^x$
 $5 = 2\sqrt{3}e^{2x}$

$\frac{5}{2\sqrt{3}} = e^{2x}$

$\log_e \left(\frac{5}{2\sqrt{3}} \right) = 2x$

$x = \frac{1}{2} \log_e \left(\frac{5}{2\sqrt{3}} \right) \approx \frac{1}{2} \log_e \frac{5\sqrt{3}}{6}$

$= 0.18$ (2dp)

$x = \frac{5}{2\sqrt{3}}$ ①

$x = \ln \frac{5\sqrt{3}}{2\sqrt{3}}$

$$6b(i) \quad \sin 2x = 2\sin^2 x \quad \text{for } 0 \leq x \leq \pi$$

$$2\sin x \cos x = 2\sin^2 x$$

$$2\sin x \cos x - 2\sin^2 x = 0$$

$$2\sin x (\cos x - \sin x) = 0$$

$$\sin x = 0 \quad \text{or} \quad \cos x - \sin x = 0$$

$$\cos x = \sin x$$

$$1 = \tan x$$

②

$\therefore$ The general solution is

$$x = n\pi + (-1)^n \left(\frac{\pi}{4}\right), \quad n\pi + \frac{\pi}{4}$$

$$= n\pi, \quad n\pi, \frac{\pi}{4}$$

ii) If $0 < x < \frac{\pi}{4}$, then

$$\sin 2x > 2\sin^2 x$$

$$\text{Consider } \sin 2x - 2\sin^2 x > 0$$

$$= 2\sin x \cos x - 2\sin^2 x > 0$$

$$= 2\sin x (\cos x - \sin x)$$

$$\text{Since } \sin x > 0 \quad \text{for } 0 < x < \frac{\pi}{4}$$

$$\text{and } \cos x > \sin x \quad \text{for } 0 < x < \frac{\pi}{4}$$

$$\text{Then } 2\sin x (\cos x - \sin x) > 0$$

$$\text{for } 0 < x < \frac{\pi}{4}$$

there are other possible methods

$$iii) \text{ Area} = \int_0^{\pi/4} y_{\text{upper}} - y_{\text{lower}} dx$$

$$= \int_0^{\pi/4} \sin 2x - 2\sin^2 x dx$$

$$= \int_0^{\pi/4} \sin 2x - (1 - \cos 2x) dx$$

$$= \left[-\frac{1}{2} \cos 2x - x + \frac{1}{2} \sin 2x \right]_0^{\pi/4}$$

$$= \left[-\frac{\pi}{4} + \frac{1}{2} \right] - \left[-\frac{1}{2} \right] = 1 - \frac{\pi}{4} \text{ m.u.}$$

Question 7

$$a) \quad P = \frac{1000}{1 + ke^{-1000t}}$$

$$i) \quad \lim_{t \rightarrow \infty} \frac{1000}{1 + ke^{-1000t}}$$

$$\text{Since } \lim_{t \rightarrow \infty} e^{-1000t} = 0$$

$$\text{then } \lim_{t \rightarrow \infty} \frac{1000}{1 + ke^{-1000t}}$$

$$= \frac{1000}{1+0}$$

$$= 1000$$

①

$$ii) \quad \text{at } t=0; P=1$$

$$P = \frac{1000}{1 + ke^0}$$

$$1 = \frac{1000}{1+k}$$

$$\therefore k = 999$$

$$\therefore P = \frac{1000}{1 + 999e^{-1000t}}$$

$$\text{when } P=500$$

$$500 = \frac{1000}{1 + 999e^{-1000t}}$$

$$2 = 1 + 999e^{-1000t}$$

$$\therefore 999e^{-1000t} = 1$$

$$e^{-1000t} = \frac{1}{999}$$

$$-1000t = \ln\left(\frac{1}{999}\right)$$

$$t \approx 2.5 \text{ days}$$

②

7(b)(i)

$$v = \frac{1}{2}(1-x^2)$$

$$\frac{dv}{dx} = -x \quad \therefore a = v \frac{dv}{dx} = \frac{x^3 - x}{2}$$

(b)(ii)

$$\frac{1}{1+x} + \frac{1}{1-x} = \frac{(1-x) + (1+x)}{(1+x)(1-x)} = \frac{2}{1-x^2}$$

$$\frac{dx}{dt} = \frac{1-x^2}{2} \Rightarrow \frac{dt}{dx} = \frac{2}{1-x^2}$$

$$\frac{dt}{dx} = \frac{1}{1+x} + \frac{1}{1-x}$$

$$t = \ln(1+x) - \ln(1-x) + c$$

$$\text{when } t=0, x=0 \quad \therefore c=0$$

$$\therefore t = \ln \frac{1+x}{1-x} \quad \text{①}$$

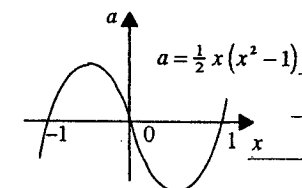
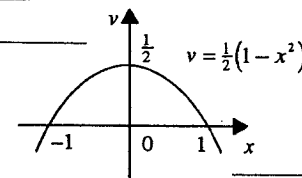
$$\frac{1+x}{1-x} = e^t$$

$$1+x = e^t - xe^t$$

$$x(e^t + 1) = e^t - 1$$

$$\therefore x = \frac{e^t - 1}{e^t + 1} = \frac{1 - e^{-t}}{1 + e^{-t}} \quad \text{②}$$

(b)(iii)



Initially the particle is at O , moving right at speed of 0.5 ms^{-1} and slowing down (since v and a have opposite signs for $0 < x < 1$).

The particle continues to move right while slowing down for $x < 1$. As $t \rightarrow \infty$, $x \rightarrow \frac{1-0}{1+0} = 1$.

Its limiting position is 1 m to the right of O .